

Health Assistant for Disease Prediction Using Hetero-Associative Memory

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Abstract. This paper presents a lightweight hybrid diagnostic framework that integrates Hetero-Associative Memory (HAM) with compact transformer-based language models, MiniLLaMA and Gemma, for early disease detection. Multilingual patient symptoms are semantically interpreted and encoded into structured vectors, which HAM processes to perform deterministic symptom–disease association and ranked prediction. By combining the conversational ability of small language models with the interpretability and low computational cost of associative memory, the proposed system enables efficient offline deployment on resource-constrained devices. Experimental results demonstrate high diagnostic performance, achieving 96.8% accuracy, 91.4% precision, 97.2% recall, and a 95.1% F1-score.

However, the dataset used in this study is internally curated and not publicly available, which limits reproducibility and independent validation of results.

Future work will focus on expanding dataset coverage and incorporating probabilistic disease modeling.

Keywords: Machine Learning, Disease Diagnosis, MiniLLaMA, Gemma, Hetero-associative Memory, Medical Chatbot, Natural Language Processing

1 Introduction

Artificial intelligence is increasingly used in healthcare for diagnosis and decision support; however, many conventional AI models rely on large datasets and computationally intensive training, limiting their use in real-time and resource-constrained settings. To address this limitation, the proposed system employs

Hetero-associative Memory (HAM), a lightweight and interpretable cognitive model that enables fast and deterministic symptom–disease mapping without repeated training, making it suitable for embedded and offline healthcare applications. To support natural and multilingual patient interaction, the framework integrates compact transformer-based language models such as MiniLLaMA and Gemma. These models enable efficient understanding of patient-reported symptoms and generation of clear, human-like medical explanations with low computational overhead.

Additionally, the system incorporates basic computer vision and autonomous navigation capabilities, allowing a humanoid robot to safely navigate hospital environments and assist patients. Together, these components form a scalable and efficient healthcare assistant that enhances accessibility and patient interaction in clinical settings. **Despite strong performance, real-world deployment requires careful consideration of dataset scalability, reproducibility, and clinical variability. The current work acknowledges these limitations and positions the system as a foundation for further development in resource-constrained healthcare environments.**

2 Literature Review

Recent advancements in healthcare AI focus on lightweight, interpretable, and on-device models for real-time deployment. Compact transformer models such as MiniLLaMA and Gemma enable multilingual medical interaction with low latency and reduced computational requirements, making them suitable for resource-constrained environments [1,2]. These models are often combined with rule-based validation mechanisms to ensure clinically safe responses [3].

Hybrid AI systems integrate language models with structured reasoning components to enhance diagnostic performance. Context-aware approaches improve symptom understanding [4], while memory-driven models support effective patient interaction in low-connectivity settings [5].

For disease prediction, Hetero-associative Memory (HAM) provides an efficient alternative to traditional machine learning models by enabling deterministic symptom–disease mapping with high interpretability and low computational cost [6]. It also supports explainable AI requirements in clinical systems [7]. Hybrid approaches combining HAM with transformer-based models further improve structured symptom extraction and disease inference [8].

Advanced approaches incorporate probabilistic extensions to handle uncertainty in diagnosis [9]. Reinforcement learning-based methods enhance adaptive reasoning in healthcare systems [10]. Deep learning techniques have been widely applied for accurate disease diagnosis in complex scenarios [11]. Additional studies demonstrate improvements using feature extraction and neural network models [12]. Intelligent healthcare data analysis methods further support scalable diagnostic frameworks [13]. IoT-based health monitoring systems enable continuous patient tracking and decision support [14]. Recent big data-driven approaches improve healthcare analytics and system efficiency [15].

3 Algorithms Implemented in the Model

3.1 Data Collection and Processing

The system collects patient symptom descriptions through text and voice inputs, which are normalized into structured representations for diagnostic prediction.

Text-Based Input Users enter free-form symptom descriptions via a web interface. Google’s Gemma 2B language model interprets conversational input, handling colloquial expressions, incomplete sentences, and non-medical terminology. The model extracts relevant symptoms, duration, and severity cues, mapping them to standardized clinical representations.

Voice Input Processing To improve accessibility, real-time speech input is supported using the browser-native Web Speech Recognition API. Spoken input is converted to text with low latency and no server-side audio processing. The system supports multilingual transcription. The transcribed text follows the same processing pipeline as text-based input.

Clinical Data Validation All medical knowledge used by the system was validated by healthcare professionals from the SIT Tumkur health center. Experts verified symptom–disease mappings, precautionary advice, specialist referrals, and patient-facing content to ensure clinical accuracy and safety.

3.2 Dataset

The dataset consists of expert-validated medical information covering **102 diseases** and **72 symptoms**. Each disease entry includes symptom associations, clinical descriptions, precautionary measures, and recommended medical specialists.

For evaluation, the dataset was split as:

- Training set: 5,916 samples (80%)
- Test set: 1,479 samples (20%)

This split supports validation of the hetero-associative memory and semantic interpretation components.

Multilingual Dataset The dataset is fully available in three languages such as English, Hindi and Kannada.

The dataset is internally curated and validated by healthcare professionals, ensuring clinical correctness. However, it is not publicly available, which limits reproducibility and independent evaluation by other researchers. Future work will focus on releasing anonymized datasets or benchmarking against publicly available datasets.

Data Quality Assurance Data quality was ensured through clinical review, evidence-based verification of precautions, multilingual translation validation by native medical professionals, and safety checks on all patient-facing recommendations. This process ensures reliable and culturally appropriate deployment in real-world healthcare environments.

3.3 System Pipeline

The system follows a structured pipeline for disease prediction. First, user input is collected in the form of text or speech. Second, MiniLLaMA or Gemma processes the input and extracts relevant symptoms. Third, the extracted symptoms are encoded into a binary symptom vector. Fourth, the vector is processed using the Hetero-associative Memory (HAM) model. Finally, the disease with the highest activation is predicted and displayed along with recommendations. The extracted symptoms from the LLM are transformed into structured binary vectors before being processed by the HAM model for final prediction.

3.4 Algorithm used for Disease Detection Model

The proposed system converts patient symptoms into structured numerical vectors and applies a Hetero-associative Memory (HAM) model for disease prediction.

MiniLLaMA MiniLLaMA is a lightweight variant of the LLaMA architecture designed for efficient text processing on resource-constrained devices. It retains the transformer-based framework while reducing model size and computational complexity, enabling faster inference with lower memory usage. As shown in Fig. 1, MiniLLaMA captures contextual relationships in input text and extracts relevant medical information for disease prediction. Its compact design makes it well suited for edge-based and offline healthcare applications.

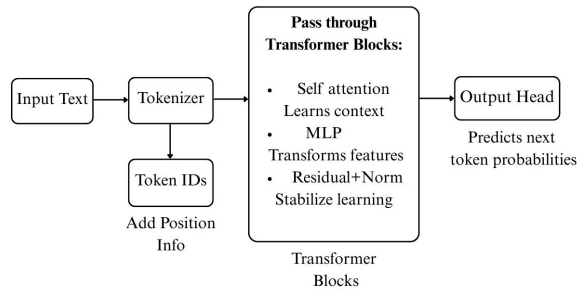


Fig. 1. Internal Architecture of MiniLLaMa.

Gemma 2B Gemma 2B is a lightweight language model designed for efficient text understanding and reasoning on resource-constrained systems. It improves semantic interpretation and multi-symptom reasoning compared to smaller models. As shown in Fig. 2, Gemma processes input text using transformer-based encoding to capture contextual relationships and support accurate disease prediction.

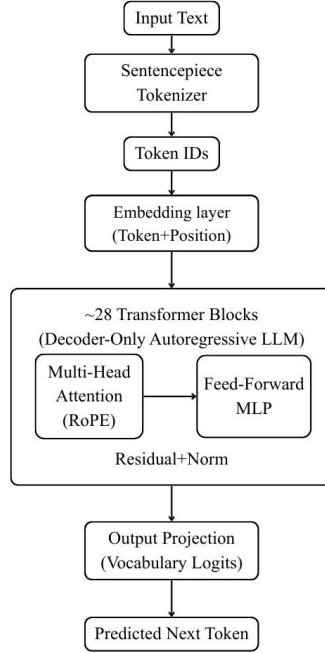


Fig. 2. Internal Architecture of Gemma 2B.

Hetero-associative Memory Model The Hetero-associative Memory (HAM) model maps input symptom vectors X to output disease vectors Y using a deterministic memory matrix. In this system, symptoms are encoded as binary vectors and diseases are represented using one-hot encoding, forming a matrix $M \in R^{72 \times 102}$.

Associations are stored during training by summing the outer products:

$$M = \sum_{k=1}^N X_k^T Y_k \quad (1)$$

During recall, a new symptom vector is used to compute the predicted output:

$$Y_{pred} = X_{input} \times M \quad (2)$$

The disease corresponding to the highest activation in Y_{pred} is selected as the final prediction. As shown in Fig. 3, this approach enables fast and interpretable disease diagnosis.

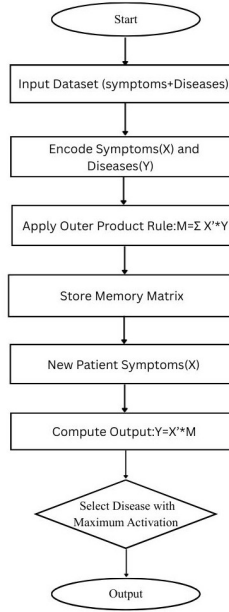


Fig. 3. Flowchart of Hetero-associative memory model

4 Model Performance

This section presents experimental results for disease prediction using MiniLLaMA and Gemma models integrated with the HAM framework.

4.1 Disease Detection Model

Both MiniLLaMA (1.1B) and Gemma-2B process natural language symptoms and convert them into structured representations for disease prediction.

Example Input: “I am suffering from headache, chills and cough.”

Predicted Disease: Common Cold

MiniLLaMA performs efficient symptom extraction suitable for resource-constrained environments, while Gemma-2B provides improved semantic understanding and multi-symptom reasoning due to its larger context capacity. The extracted symptoms are mapped into structured vectors and processed by the HAM model to generate the final prediction.

Performance Analysis

As shown in Table 1, MiniLLaMA processes inputs in smaller chunks, resulting in higher processing time but enabling deployment on low-power devices such as the Raspberry Pi 4. In contrast, Gemma-2B processes the input in a single chunk, yielding higher accuracy and confidence scores, but requires greater computational resources and is unsuitable for edge deployment. Overall, Gemma-2B offers superior reasoning performance, while MiniLLaMA is better suited for resource-constrained environments.

Table 1. Performance Comparison of MiniLLaMA and Gemma-2B

Parameter	MiniLLaMA (1.1B)	Gemma-2B
Model Size	1.1B parameters	2B parameters
Context Window	512 tokens	2048–4096 tokens
Chunk Size	8 tokens/chunk	Single token sequence
Number of Chunks	2	1
CPU Processing Time	1.6 sec	1.94 sec
GPU Processing Time	0.91 sec	0.67 sec
Symptom Accuracy	89%	94%
Multi-symptom Reasoning	Medium	High
Predicted Disease	Common Cold	Common Cold
Disease Confidence	84%	91%

The HAM model associates patient symptom vectors with disease vectors, enabling reliable recall even with partial or noisy inputs. The system supports **102 disease patterns**, with symptoms encoded as input vectors and linked to disease outputs using Hebbian learning. Disease labels were validated against clinically verified ground truth data. Performance analysis for a subset of seven diseases is presented in Table 2.

Table 2. Confusion Matrix for Disease Classification

A / P	Vertigo	AIDS	Acne	Hepatitis	Allergy	Arthritis	Asthma
Vertigo	104	2	0	0	0	5	2
AIDS	0	110	0	0	2	0	3
Acne	0	2	114	2	0	0	0
Hepatitis	5	0	0	105	3	3	0
Allergy	2	0	0	3	106	3	0
Arthritis	0	0	0	1	1	113	1
Asthma	2	0	2	0	2	0	112

4.2 Performance Comparison of Algorithms

Model performance was evaluated against clinically verified ground truth data. As shown in Table 3, disease prediction accuracy reached 85% for Decision Trees, 90.4% for KNN, and 92.1% for SVM. The Hetero-associative Memory (HAM) model achieved the highest accuracy of 96.8%, demonstrating strong capability in disease grouping and prediction. KNN performance benefited from probability-based similarity measures, while SVM leveraged hyperplane-based classification. HAM further enabled stratified and explainable diagnosis through weight-based associative mapping.

Table 3. Evaluation Metrics Across Algorithms

Metrics	HAM	SVM	KNN	Decision Tree
Accuracy (%)	96.8	92.1	90.4	85.0
Precision (%)	91.4	88.5	87.3	80.0
Recall (%)	97.2	91.2	88.9	85.0
F1-Score (%)	95.1	90.8	89.1	82.0

4.3 Inference

To analyze misclassification behavior, two representative cases were examined: *Allergy vs Asthma* and *Vertigo vs Arthritis*. In both cases, confusion arises due to overlapping or non-specific symptoms. Traditional models such as Decision Tree and KNN show higher misclassification when symptom boundaries are unclear or incomplete. SVM reduces this effect through non-linear decision boundaries, while the Hetero-associative Memory (HAM) model further improves recall by leveraging strong associative symptom–disease mappings. Overall, HAM demonstrates greater robustness in handling ambiguous symptom patterns compared to classical classifiers.

Misclassification can occur due to overlapping symptoms between diseases such as Allergy and Asthma, or Vertigo and Arthritis. To address this limitation, future work will incorporate probabilistic and weighted associative memory models, enabling confidence-based prediction and improved robustness.

5 Limitations and Future Work

The system is limited by dataset size, binary symptom encoding, lack of public data availability, and absence of statistical validation. Future work will address these issues by expanding disease coverage, introducing probabilistic models, evaluating multilingual consistency, and validating the system on real-world datasets.

6 Conclusion

This study demonstrates the effectiveness of Hetero-associative Memory (HAM) for interpretable and efficient disease diagnosis through symptom–disease mapping. The integration of lightweight language models such as MiniLLaMA and Gemma enables multilingual interaction and deployment in resource-constrained environments.

Despite high accuracy, limitations include dataset size, lack of public availability, and absence of statistical validation. Future work will focus on expanding disease coverage, incorporating probabilistic models, and validating performance on real-world datasets.

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Ethics and Availability Statements

Ethics approval and consent to participate: Not applicable.

Consent for publication: Not applicable.

Data availability: Available from the corresponding author on reasonable request.

Code availability: Available from the corresponding author on reasonable request.

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